MINI PROJECT

ABSTRACT

Department of Artificial Intelligence and Machine Learning

FreshAir Desk

AI-Powered Smart Desk Environment Monitor

The quality of the surrounding environment significantly influences an individual's health, comfort, concentration, and productivity, particularly for students and professionals who spend extended hours studying or working indoors. Environmental factors such as poor air quality, inadequate lighting, uncomfortable temperature, excessive humidity, and high noise levels often go unnoticed but can lead to fatigue, headaches, reduced focus, and decreased work efficiency.The proposed project, FreshAir Desk, aims to design and develop a compact IoT-based smart device capable of continuously monitoring the environmental conditions around a study table or office desk. The system employs an ESP32 microcontroller integrated with multiple sensors to measure essential environmental parameters, including temperature, humidity, light intensity, air quality (VOC/eCO2 or CO2), and optional noise levels. The collected data is processed in real time and transmitted to a web or mobile dashboard for visualization and analysis.Unlike conventional environmental monitoring systems that simply display sensor readings, the proposed solution incorporates an Artificial Intelligence (AI) or rule-based analytical model to evaluate the collected data and generate meaningful indicators such as Comfort Score, Focus Score, and Air Quality Score. Based on these assessments, the system provides personalized recommendations such as improving ventilation, adjusting lighting, regulating temperature, minimizing noise, and taking short breaks when conditions become unfavorable. The project integrates IoT, Embedded Systems, Environmental Sensing, Artificial Intelligence, and Cloud Connectivity into a unified smart monitoring platform, enabling proactive decision-making rather than merely reporting environmental conditions. FreshAir Desk demonstrates how environmental monitoring combined with intelligent recommendations can enhance indoor comfort, improve productivity, and contribute to better health and well-being in modern learning and working environments.

Keywords: Internet of Things, Artificial Intelligence, Environmental Monitoring, ESP32, Smart Desk, Air Quality, Embedded Systems.

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